

Not All Information Brings Benefits: Personalization-Driven Agent Debate for Conversational Recommendation

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Abstract

Conversational recommender systems (CRSs) aim to provide real-time recommendations through dynamic interactions between users and the system. Recent studies have revealed the value of personalized information derived from users' historical dialogue records in refining user preferences. However, existing methods often utilize the entire historical dialogue of a user indiscriminately, leading to the issue of **cognitive negative transfer**, wherein historical dialogue sessions impede rather than facilitate current decision-making. This ultimately degrades the performance of conversational recommendations.

To address this issue, inspired by the behavioral decision theory, this paper proposes a novel model, Personalization-Driven Agent Debate for Conversational Recommendation, named **EyeCRS**. EyeCRS is comprised of two core components: (i) The *Multi-Agent Debate Module* employs supporting and opposing agents to simulate a human-like debate process, thereby enabling a rigorous evaluation of whether historical dialogue sessions may introduce cognitive negative transfer into the recommender. (ii) The *Parametric Injection Module*, subsequently, parameterizes the historical dialogues that are validated as positive and injects them into the intrinsic parameter space of the downstream recommender, thereby effectively enhancing the model's ability to internalize and utilize personalized knowledge. Experimental results on two real-world datasets show that EyeCRS effectively mitigates cognitive negative transfer and achieves superior performance in conversational recommendation.

CCS Concepts

• Information systems → Recommender systems.

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Conversational Recommendation, Multi-Agent Debate, Large Language Models

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1 Introduction

Conversational Recommender Systems (CRSs), recommending items to users through multi-round natural language dialogues, have garnered significant attention in recent years due to their capacity to facilitate dynamic interactions between users and systems [18, 23, 33, 55, 57]. This interaction enables real-time refinement of user needs and feedback, resulting in more personalized and accurate recommendations [20, 44, 54, 56, 58].

Early CRSs [41, 52, 53] primarily focused on modeling user preferences based solely on the current conversation. Despite the success of early exploration, they often ignore personalized information derived from users' historical dialogues, resulting in the inability to effectively capture and adapt to the evolving nature of user preferences. Fortunately, recent research [22, 39] has acknowledged this limitation and introduced conversational recommendation methods that integrate personalized information by leveraging users' historical dialogues to refine and enhance recommendations during current interactions.

However, existing conversational recommendation methods generally adopt an 'irrigation strategy' for handling personalized information, in which all past dialogue sessions are indiscriminately incorporated into the recommendation process. This often leads to cognitive negative transfer, a concept from cognitive psychology that describes the erroneous application of prior knowledge or skills in new contexts, thereby impairing learning and performance [38]. In the context of CRS, cognitive negative transfer refers to the phenomenon where invalid or misleading information from historical dialogues interferes with the system's cognitive processes, thereby undermining its decision-making capabilities.

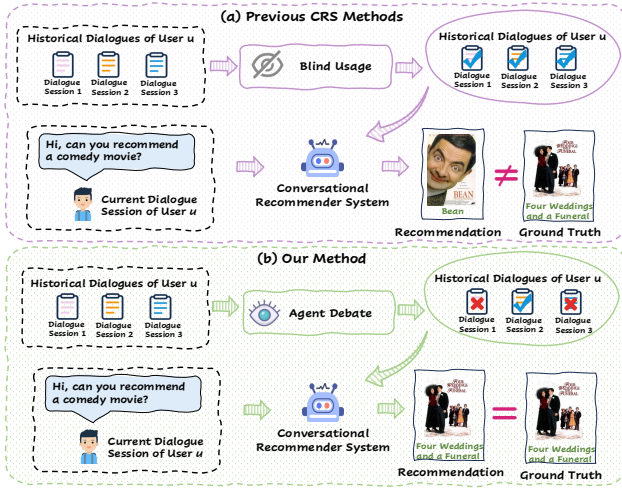


Figure 1: Previous CRS methods vs. our method.

As illustrated in Figure 1 (a), if personalized information from historical dialogues is indiscriminately utilized, the system may not only fail to benefit from it but may also introduce the problem of cognitive negative transfer, thereby undermining recommendation performance. To further validate this problem, we conducted empirical experiments and the results are shown in Figure 2.¹ Observe from Figure 2, blindly incorporating historical dialogue information yields marginal gains in a small subset of samples (nearly 1/3), but in most cases it results in no improvement or even negative effects. These findings provide strong evidence that cognitive negative transfer indeed exists in conversational recommendation tasks and significantly constrains the improvement of recommendation performance.

To address this issue, we take inspiration from behavioral decision theory, which suggests that humans must assess whether past knowledge remains applicable to the current context [28]. We argue that CRSs should also possess situational perception and information discrimination to perform better recommendations. As shown in Figure 1 (b) of the illustration of our method, if the system can distinguish between beneficial and detrimental personalized information from historical dialogues, it can produce recommendations that better align with the user’s immediate intent.

Although we highlight the potential of such “human-like” situational awareness in conversational recommendation, there are still two core challenges to address: (i) How to introduce situational awareness to distinguish useful and harmful personalized information in historical sessions? (ii) How to strategically leverage useful historical dialogues so that effective signals benefit current recommendations? To tackle the aforementioned challenges, we propose a novel model of Personalization-Driven Agent Debate for Conversational Recommendation (**EyeCRS**). The model is equipped with two novel modules, the Multi-Agent Debate Module and the Parametric Injection Module. In the Multi-Agent Debate Module, we draw inspiration from the notion that “the truth becomes clearer

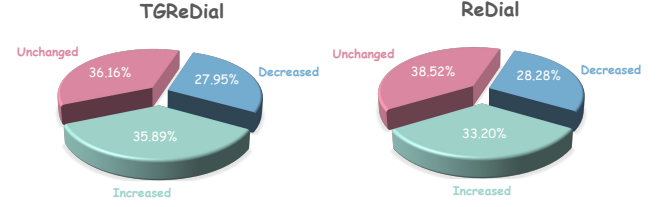


Figure 2: Distribution of cognitive negative transfer phenomena on TGRReDial and ReDial datasets.

through debate” [24]. Specifically, given a user’s historical dialogue sessions, the model constructs supporter and opponent agents from opposing perspectives to articulate and challenge the usefulness of the historical information. Subsequently, a judge agent is introduced to synthesize the debate and deliver a more reliable and objective conclusion. In the Parametric Injection Module, we first parameterize the historical dialogues that are validated as positive through the debate stage. These parameterized representations are then injected into the large language model (LLM)-based recommender, transforming personalized information into internalized knowledge and enhancing the model’s ability to perceive user interests.

We summarize the contributions of this work as follows:

- We highlight the issue of cognitive negative transfer and emphasize its importance in conversational recommendation.
- We propose EyeCRS, a novel CRS that integrates the multi-agent debate and the parametric injection to autonomously identify and leverage personalized historical dialogues, thereby alleviating the problem of knowledge negative transfer and improving CRSs.
- Experimental results on two benchmark datasets demonstrate that our proposed model effectively mitigates the issue of cognitive negative transfer and outperforms state-of-the-art CRS methods.

To the best of our knowledge, this paper is the first effort to highlight the issue of cognitive negative transfer in CRS and provide an effective solution to mitigate it.

2 Related Work

2.1 Conversational Recommender Systems

We first review the key literature on CRSs, categorizing them into two types based on whether user historical dialogue sessions are incorporated: general CRSs [6, 17, 25, 27, 42] and CRSs with personalized historical dialogues [22, 39]. **General CRSs.** Early CRSs [7, 52, 59] primarily relied on knowledge graph-based methods to capture semantic relations, enabling the system to generate recommendations. Subsequently, models such as UniCRS [34], DCRS [8], and DisenCRS [3], incorporate the DialoGPT [49] language model to better capture conversational context and user intent. Recently, the capabilities of LLMs have gained widespread recognition [19, 26, 40, 46], and their application in CRSs has gradually become a research hotspot. Specifically, He et al. [13] provide a comprehensive investigation into the capabilities of LLMs for zero-shot conversational recommendation. Yang and Chen [43] propose ReFICR, a retrieval-enhanced CRS that jointly optimizes retrieval and generation within LLMs. Wang et al. [33] propose an LLM-driven interactive evaluation framework that employs simulated

¹Using the UCCR [22] model as the backbone, we classify each sample based on the change in similarity to the ground truth after incorporating personalized historical dialogues. Samples with a similarity gain greater than 0.1 are labeled *increased*, those with a decrease greater than 0.1 are labeled *decreased*, and the rest are labeled *unchanged*.

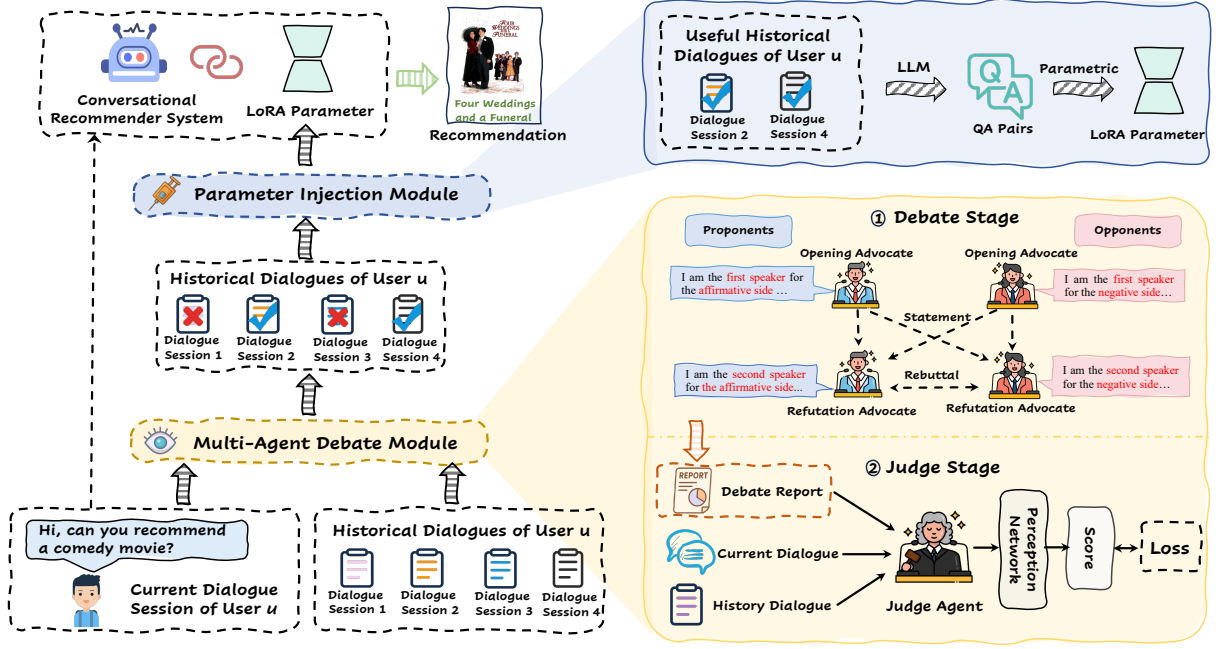


Figure 3: Overview of EyeCRS and its two components: (i) Multi-Agent Debate Module, which filters out unhelpful historical dialogue sessions to mitigate cognitive negative transfer; and (ii) Parametric Injection Module, which injects beneficial historical knowledge into the intrinsic parameter space of the LLM for personalized reasoning.

users to assess dialogue quality. Together, these studies facilitate the rapid evolution of LLM-based CRSs.

CRSs with Personalized Historical Dialogues. CRSs with personalized historical dialogues aim to provide personalized recommendations tailored to each individual by utilizing users’ historical dialogue sessions. Research in this area is still in its early stages, with limited studies available. Li et al. [22] propose the first work to emphasize the importance of user historical dialogues as a crucial source of information for CRSs. Subsequently, Xi et al. [39] introduce a memory-augmented CRS framework, powered by LLMs, to manage users’ historical dialogue sessions. However, they suffer from the problem of cognitive negative transfer. In contrast, our proposed EyeCRS model focuses on when to use historical dialogues and which historical dialogues to leverage for recommendations, thus mitigating cognitive negative transfer and integrating personalized historical dialogues.

2.2 LLM-Based Multi-Agent Debate

The concept of multi-agent debate originates from Irving et al. [16], which introduces adversarial argumentation among multiple agents, with a judge selecting the more persuasive side to improve interpretability and correctness. With the rapid advancement of LLMs, Du et al. [10] demonstrate that multi-agent debate can significantly enhance factual accuracy and complex reasoning in LLMs. Building on this idea, Chan et al. [5] explore the ChatEval framework, showing that communication strategies and role diversity among agents influence evaluation accuracy. Furthermore, Estornell and Liu [12] analyze the dynamics of multi-agent debate, identifying issues such as ‘tyranny of the majority’ and ‘shared misconceptions’, and propose debate pruning to improve stability and interpretability. Different from these studies, we are the first, to the best of our

knowledge, to introduce multi-agent debate into the field of CRSs. By leveraging the debate mechanism to distinguish between beneficial and detrimental historical dialogues, our approach effectively alleviates cognitive negative transfer and enhances CRS robustness.

3 Method

Problem Definition. Let $\mathcal{U}$ and $\mathcal{I}$ denote the user set and item set in the system, respectively. For a user $u \in \mathcal{U}$, the *current dialogue session* C represents the current dialogue, in which the system provides recommendations. All complete dialogues associated with user u prior to dialogue C are denoted as *historical dialogue sessions*, represented by $\mathcal{H}_u = \{h_i\}_{i=1}^{n_u}$, where n_u denotes the number of historical dialogues associated with user u .²

Based on the above definitions, CRSs aim to (i) recommend a subset $\mathcal{I}_{\text{rec}} \subseteq \mathcal{I}$ that best matches the user’s intent expressed in the current dialogue session C and (ii) to generate relevant and coherent responses regarding these recommended items.

3.1 Overview of EyeCRS

Existing models often treat all historical dialogues as equally useful, which may cause cognitive negative transfer—where irrelevant or outdated preferences distort the understanding of the current intent. To address this issue, we propose a refined CRS framework, EyeCRS, that selectively extracts a beneficial subset $\mathcal{H}_u^{\text{use}} \subseteq \mathcal{H}_u$ instead of leveraging all past sessions indiscriminately. An overview of our proposed EyeCRS model is presented in Figure 3. Specifically, we introduce two sequential modules to achieve our goal:

²Following previous work [22, 39], we distinguish *historical dialogue sessions* from the commonly used term *dialogue history* in most CRSs [7, 8, 34, 52]. The latter refers to preceding utterances within the same session (turn-level), whereas the former denotes complete past sessions occurring before the current one.

Multi-Agent Debate Module assesses the relevance of each historical dialogue to the current conversation and preserves the helpful historical dialogue sessions through a structured multi-agent debate (see Section 3.2).

Parametric Injection Module transforms beneficial historical dialogue sessions into the intrinsic parameter space, and aligns them with the downstream LLM to perform recommendations (see Section 3.3).

For recommendation, the current dialogue session is fed into an LLM-based recommender that incorporates user-specific parameters from the Parametric Injection Module, enabling personalized item prediction based on the injected user preferences. For response generation, the system invokes an external LLM conditioned on the current dialogue session, the recommended items, and the selected historical dialogues identified by the Multi-Agent Debate Module to generate contextually appropriate and personalized responses.

3.2 Multi-Agent Debate Module

Building on the overall framework, the Multi-Agent Debate Module aims to identify which historical dialogue sessions are truly relevant to the current context. It achieves this through structured debates among multiple LLM-based agents, whose contrasting views provide evidence for evaluating the usefulness of each historical session. The following describes how the debate process is organized and how relevance is quantified.

3.2.1 Debate Organization. For each historical session h_i from the user's dialogue historical sessions $\mathcal{H}_u$, a debate is organized together with the current session C to assess whether h_i provides beneficial cognitive cues or causes cognitive interference. There are two opposing teams:

- **Proponent** ($Stance_{pro}$): argues that h_i offers valuable prior knowledge that helps interpret the user's current intent.
- **Opponent** ($Stance_{opp}$): contends that h_i may induce cognitive negative transfer.

These two teams engage in a two-stage debate—*Statement* and *Rebuttal*—on the above motion,³ as illustrated below.

In the *Statement* stage, each agent independently generates an initial argument based on the current session C , the historical session h_i , and its assigned stance. Let $j \in \{pro, opp\}$ index the two agents. The statement text from the j -th agent is defined as:

$$d_j^{(1)} = f_{stnt}(C, h_i, Stance_j), \quad (1)$$

where $f_{stnt}(\cdot)$ denotes a text-generation function implemented via an LLM call with a structured prompt.⁴

In the *Rebuttal* stage, each agent reviews the opponent's statement and generates counterarguments accordingly. The rebuttal from the j -th agent is defined as:

$$d_j^{(2)} = f_{reb}(C, h_i, D_i^{(1)}, Stance_j), \quad (2)$$

where $f_{reb}(\cdot)$ denotes an LLM-based text generation function for rebuttal,⁵ and $D_i^{(1)}$ represents the collection of statements produced in the previous stage.

³Inspired by the Lincoln–Douglas debate format and recent advances in multi-agent prompting [9, 11, 24].

⁴The detailed prompt used in this stage is provided in Appendix A.1.

⁵The detailed prompt used in this stage is provided in Appendix A.2.

The complete debate record for a historical session h_i is:

$$\mathcal{D}_i = \{D_i^{(1)}, D_i^{(2)}\}, \quad (3)$$

which captures the full exchange between the two agents. These debate traces $\mathcal{D}_i$ serve as input for relevance qualification described in the next subsection.

3.2.2 Relevance Qualification. To quantitatively evaluate the outcomes of the debate, we introduce a lightweight *judge* model that scores the relevance of each historical dialogue session with respect to the current context. The judge is built upon a fixed LLM encoder, *Qwen3-Embedding*—a variant of the Qwen3 model fine-tuned for dense representation learning [31, 35, 48].

Given a current dialogue session C , a historical dialogue h_i , and its debate log $\mathcal{D}_i$, we construct a natural language prompt $\mathcal{P}_i^{\text{judge}}$ by concatenating them using a predefined textual template. The prompt $\mathcal{P}_i^{\text{judge}}$ is then fed into the encoder, and we obtain the dense representation $\mathbf{z}_i \in \mathbb{R}^m$ from the hidden state corresponding to the end-of-sequence token:

$$\mathbf{z}_i = \text{Encoder}^{\text{judge}}(\mathcal{P}_i^{\text{judge}})[[\text{EOS}]]. \quad (4)$$

To transform this representation into a scalar relevance estimate, $\mathbf{z}_i$ is passed through a multilayer perceptron (MLP) scoring head to compute a relevance score s_i :

$$s_i = \sigma(\mathbf{w}_2^\top \text{ReLU}(\mathbf{W}_1 \mathbf{z}_i + \mathbf{b}_1) + b_2). \quad (5)$$

The score s_i quantifies the cognitive influence of each historical session h_i on the current dialogue C , reflecting how beneficial or interfering the session is to the current recommendation context. For each user u , the judge is applied to all historical sessions in $\mathcal{H}_u$, and those with scores higher than a threshold τ are retained to form the cognitive-guided subset $\mathcal{H}_u^{\text{use}}$.

As explicit annotations of cognitive influence are unavailable, we train the judge model using weak supervision signals [36] derived from recommendation outcomes. For each current dialogue C and its historical candidates $\mathcal{H}_u = \{h_i\}_{i=1}^{n_u}$, we measure how incorporating each h_i affects the model's similarity between the dialogue representation and the ground-truth item embedding $\mathbf{v}_g$, where $\mathbf{v}_g$ is encoded by the recommender model's item encoder. Let $\text{sim}(\cdot, \cdot)$ denote the cosine similarity, and define the relative gain of h_i as:

$$\Delta_i = \text{sim}(\text{Encoder}_\theta^{\text{rec}}(C, h_i), \mathbf{v}_g) - \text{sim}(\text{Encoder}_\theta^{\text{rec}}(C), \mathbf{v}_g), \quad (6)$$

where $\text{Encoder}_\theta^{\text{rec}}(\cdot)$ denotes the session encoder built upon a pre-trained LLM with frozen base parameters θ . Sessions with $\Delta_i > 0$ (i.e., improving item similarity) are treated as positive examples h^+ , while those with $\Delta_i < 0$ are treated as negative examples h^- . These pseudo-labels provide weak supervision for optimizing the judge.

Given the corresponding debate-informed prompts $\mathcal{P}_{h^+}^{\text{judge}}$ and $\mathcal{P}_{h^-}^{\text{judge}}$, the judge produces scores s^+ and s^- according to Eq. (5). The model is then optimized with a pairwise contrastive loss that encourages higher scores for cognitively helpful histories:

$$\mathcal{L}_{\text{judge}} = -\log \sigma(s^+ - s^-). \quad (7)$$

During training, the encoder parameters remain frozen, and only the scoring head is updated. This enables the judge to learn to identify cognitively beneficial sessions from weak supervision without requiring manual labels.

3.3 Parametric Injection Module

In this section, we present the parametric injection module, which embeds the selected historical dialogues into the model through parameter-efficient adaptation. For each user, the retained dialogues are transformed into question-answer (QA) pairs that represent individualized cognitive preferences. These QA pairs are used to train a lightweight Low-Rank Adaptation (LoRA) [15] adapter specific to that user, which captures long-term cognition and is dynamically loaded at inference for personalized recommendations.

3.3.1 QA-Based Data Generation. As the first step of parameter injection, we prompt an instruction-tuned LLM to synthesize preference-focused QA pairs from each selected historical dialogue, rather than extracting verbatim spans. This makes the user's implicit preferences explicit and model-consumable.

Formally, for each $h_i \in \mathcal{H}_u^{\text{use}}$, we apply an LLM-based QA generation function $f_{\text{QA}}(\cdot)$ with a predefined prompt template to produce a set of QA pairs:

$$\mathcal{Q}_i = f_{\text{QA}}(h_i) = \{(q_{ip}, a_{ip})\}_{p=1}^{n_{\text{QA}}}, \quad (8)$$

where q_{ip} is a synthesized question that captures a specific aspect of the user's preference implied by h_i (e.g., genre, tone, actor affinity), and a_{ip} is the corresponding answer grounded in the dialogue context. Each h_i yields multiple QA pairs to cover diverse aspects of long-term user cognition.

We further aggregate all QA pairs derived from the selected historical dialogues into a set $\mathcal{Q}_u$, which serves as user-level supervision for fine-tuning user-specific adapters in the next stage:

$$\mathcal{Q}_u = \bigcup_{h_i \in \mathcal{H}_u^{\text{use}}} \mathcal{Q}_i. \quad (9)$$

3.3.2 User-Level LoRA Adaptation. Using the aggregated QA pairs $\mathcal{Q}_u$, we follow the standard LoRA design [15] to fine-tune a user-specific adapter for each user. Here, LoRA is applied to the attention projection layers of the encoder, where a pair of low-rank trainable matrices is injected to enable parameter-efficient personalization.

To implement this personalization, the user-specific adapter ϕ_u is trained using the aggregated QA supervision $\mathcal{Q}_u$. Each QA pair is converted into a token sequence:

$$x_{ip} = [q_{ip} \oplus a_{ip}], \quad (10)$$

where $\oplus$ denotes sequence concatenation. During training, the model predicts only the answer tokens a_{ip} while treating the question part as context.

We then optimize the LoRA-adapted parameters ϕ_u with a causal language modeling objective while keeping the base model θ frozen:

$$\min_{\phi_u} \sum_{(q_{ip}, a_{ip}) \in \mathcal{Q}_u} \sum_{t=t_{\text{start}}(a_{ip})}^{n_{ip}} -\log P_{\theta, \phi_u}(x_{ip,t} \mid x_{ip,<t}), \quad (11)$$

where $t_{\text{start}}(a_{ip})$ denotes the index of the first token in the answer segment a_{ip} , and n_{ip} is the total length of the concatenated input x_{ip} .

3.4 Recommendation

The recommendation module unifies general collaborative knowledge and user-specific personalization. During training, it learns a global dialogue-item matching function, while personalization at

inference is achieved by injecting a user-specific adapter ϕ_u from the parametric injection stage.

3.4.1 Training for Recommendation. To learn general dialogue-item matching patterns, the model is trained without any user-specific adapters. For each current dialogue C , we construct a recommendation prompt $\mathcal{P}^{\text{rec}}$ using a predefined template and feed it into the frozen base encoder Encoder_{θ} to obtain a query representation:

$$\mathbf{q} = \text{Encoder}_{\theta}^{\text{rec}}(\mathcal{P}^{\text{rec}})[[\text{EOS}]], \quad (12)$$

where θ denotes the base model parameters.

Let $\mathbf{E}_I = [\mathbf{v}_1, \dots, \mathbf{v}_N]^T \in \mathbb{R}^{N \times m}$ be the embedding matrix of N candidate items. The matching score between a query $\mathbf{q}$ and an item $\mathbf{v}_c$ is computed using a decomposed bilinear function:

$$r_c = \boldsymbol{\alpha}^T \mathbf{q} + \boldsymbol{\beta}^T \mathbf{v}_c + (\boldsymbol{\gamma} \odot \mathbf{q})^T \mathbf{v}_c + b, \quad (13)$$

where $\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma} \in \mathbb{R}^m$ and $b \in \mathbb{R}$ are trainable parameters, and $\odot$ denotes element-wise multiplication.

We adopt a full softmax objective over the entire item corpus. For each training dialogue C , the model predicts a ranking distribution over all items:

$$\mathcal{L}_{\text{rec}} = -\log \frac{\exp(r_g)}{\sum_{c=1}^N \exp(r_c)}, \quad (14)$$

where g denotes the ground-truth item associated with the dialogue C . This objective encourages the encoder to learn user-independent, collaborative matching patterns.

3.4.2 Inference for Recommendation. At inference time, the key difference from training is that personalization is introduced by activating the corresponding user adapter ϕ_u . The adapted encoder $\text{Encoder}_{\theta, \phi_u}^{\text{rec}}$ encodes the same recommendation prompt $\mathcal{P}^{\text{rec}}$ to produce a personalized query embedding:

$$\mathbf{q}_u = \text{Encoder}_{\theta, \phi_u}^{\text{rec}}(\mathcal{P}^{\text{rec}})[[\text{EOS}]]. \quad (15)$$

Except for this personalization step, the inference process follows the same procedure as in training.

4 Experiments

4.1 Experimental Setup

4.1.1 Datasets. We conduct experiments on two public datasets, TGRDial and ReDial, following previous work [32, 50]. ReDial [21] is a movie-related conversational dataset that was constructed through crowdsourcing on Amazon Mechanical Turk (AMT). It consists of 10,006 dialogues with a total of 182,150 utterances, covering 51,699 movies and involving 504 users. TGRDial [53] is a Chinese movie-based CRSs dataset. It contains 10,000 sessions with 129,392 utterances, involving 1,482 users and 33,834 movies.

4.1.2 Evaluation Metrics. For the recommendation task, we adopt several widely used evaluation metrics, including HR@K, NDCG@K, and MRR@K, as done in An et al. [2], Ao et al. [4], He et al. [14]. These metrics help assess the performance of the recommendation task, where K is chosen from $\{5, 10, 20\}$. For the response generation task, we follow existing studies that use human evaluation [3, 37, 39], where three annotators score the generated responses based on fluency and informativeness. Each annotator's score ranges from 0 to 3, and the final result is obtained by averaging the scores of the three annotators.

Table 1: Performance comparison of different models on the recommendation task. The best results are boldface, while the second-best results are underlined. The symbol * denotes statistically significant improvements over the best baseline at $p < 0.05$.

Model	TGRReDial									ReDial								
	HR			MRR			NDCG			HR			MRR			NDCG		
	@5	@10	@20	@5	@10	@20	@5	@10	@20	@5	@10	@20	@5	@10	@20	@5	@10	@20
ReDial	0.0030	0.0055	0.0102	0.0015	0.0018	0.0021	0.0018	0.0027	0.0038	0.0293	0.0413	0.0882	0.0174	0.0190	0.0223	0.0203	0.0242	0.0361
KBRD	0.0050	0.0112	0.0174	0.0026	0.0035	0.0040	0.0032	0.0053	0.0069	0.0824	0.1395	0.2185	0.0337	0.0418	0.0470	0.0457	0.0646	0.0842
KGSF	0.0112	0.0149	0.0249	0.0039	0.0044	0.0051	0.0057	0.0068	0.0094	0.0908	0.1378	0.2319	0.0385	0.0445	0.0508	0.0514	0.0663	0.0898
TGRReDial	0.0075	0.0174	0.0236	0.0055	0.0068	0.0072	0.0059	0.0091	0.0107	0.0874	0.1395	0.2336	0.0433	0.0502	0.0567	0.0543	0.0710	0.0948
UCCR	0.0087	0.0174	0.0286	0.0071	0.0082	0.0090	0.0075	0.0103	0.0131	0.1059	0.1782	0.2672	0.0443	0.0538	0.0596	0.0594	0.0826	0.1047
UniCRS	0.0050	0.0124	0.0236	0.0024	0.0035	0.0042	0.0031	0.0055	0.0083	0.1005	0.1605	0.2480	0.0392	0.0467	0.0528	0.0542	0.0731	0.0952
ZSCRS	0.0025	0.0087	0.0100	0.0007	0.0016	0.0017	0.0012	0.0032	0.0035	0.1261	0.1882	0.2353	0.0511	0.0594	0.0629	0.0695	0.0896	0.1018
MemoCRS	0.0162	0.0261	0.0323	0.0095	0.0108	0.0112	0.0111	0.0143	0.0158	0.1361	0.2151	0.2857	0.0718	0.0821	0.0871	0.0875	0.1128	0.1308
EyeCRS	0.0191*	0.0282*	0.0489*	0.0097*	0.0111*	0.0123*	0.0120*	0.0151*	0.0205*	0.1510*	0.2273*	0.3215*	0.0806*	0.0906*	0.0970*	0.0980*	0.1225*	0.1461*

4.1.3 Baselines. We apply the following representative baselines for comparison: (1) **ReDial** [21] proposes a dialogue module based on HRED [29] and a recommendation module based on an auto-encoder. (2) **KBRD** [7] integrates knowledge graphs into CRSs, linking recommendation and dialogue systems through knowledge propagation. (3) **KGSF** [52] combines word-oriented and entity-oriented knowledge graphs to improve data representation in CRSs. (4) **TGRReDial** [53] is proposed with the TGRReDial dataset and incorporates a topic prediction task [60] to enhance performance. (5) **UCCR** [22] jointly models current dialogue sessions, historical dialogue sessions, and similar users in a user-centric manner. (6) **UniCRS** [34] pioneers the unification of recommendation and dialogue tasks within the prompt learning paradigm. (7) **Zero-Shot CRS** [13], abbreviated as ZSCRS, is the first to employ LLMs as zero-shot conversational recommenders. (8) **MemoCRS** [39] introduces a memory-augmented CRS framework, powered by LLMs, to manage users' historical dialogues.

4.1.4 Implementation Details. We follow the experimental setup of MemoCRS [39]. Apart from LLM-based methods, all other baselines are implemented using the open-source toolkit CRSLab [51]. Methods based on closed-source LLMs, including Zero-shot LLM [13] and MemoCRS [39], utilize GPT-4 [1] as the backbone for recommendation generation. Our model also adopts GPT-4 for debate and QA generation, while using open-source LLMs for other components: Qwen3-Embedding-4B for the judge and Qwen3-Embedding-8B for the recommender. In data processing, we follow Xi et al. [39] and re-split the datasets into training, validation, and test sets in chronological order. A random subset of users is selected, with their most recent conversations used for validation and test sets and the remaining ones for training. To avoid shortcuts caused by repeated entries [13], we filter out dialogues where recommended items have appeared earlier.

4.2 Overall Performance

To validate the effectiveness of our proposed EyeCRS, we conduct comprehensive comparisons with several competitive state-of-the-art baselines in CRSs, evaluating performance on both recommendation tasks and response generation tasks.

Evaluation on Recommendation Task. The comparison of recommendation performance between our model and the baselines, is shown in Table 1. We have the following observations:

- (1) Our proposed EyeCRS significantly outperforms existing baselines, fully demonstrating its superior recommendation performance. Specifically, on the TGRReDial and ReDial datasets,

EyeCRS achieves substantial improvements in the average values of HR, MRR, and NDCG at ranking list lengths of @5, @10, and @20. Together, these results consistently validate the effectiveness and superiority of the EyeCRS model in CRSs.

- (2) Methods that incorporate personalized information (e.g., UCCR, MemoCRS) generally outperform general conventional methods. This observation underscores the pivotal value of personalized information contained in users' historical dialogues for modeling user interests and highlights its crucial role in enhancing the effectiveness of CRSs.
- (3) EyeCRS significantly outperforms existing methods that leverage personalized information from historical dialogues, such as UCCR and MemoCRS, across multiple key evaluation metrics. This highlights the superiority of EyeCRS in equipping CRSs with self-perception and discriminative capabilities, in contrast to traditional methods that blindly use historical dialogues. By employing a multi-agent debate mechanism to selectively filter historical information and integrating it through parametric injection, EyeCRS effectively mitigates the adverse effects of cognitive negative transfer, thereby achieving substantial improvements in recommendation performance.

Evaluation on Response Generation Task. To evaluate the performance of our proposed method in response generation, we follow the evaluation settings of Zhang et al. [47] and Xi et al. [39], conducting both automatic evaluation using GPT-4 and human annotation from two aspects: Fluency and Informativeness.

As shown in Table 2, the traditional conversational recommendation models such as KBRD, KGSF, and UniCRS perform significantly worse than LLM-based methods such as ZSCRS and MemoCRS on both Fluency and Informativeness. This result indicates that the prior advantages of LLMs in language generation and knowledge integration can be directly translated into more natural, coherent, and informative responses. Secondly, models that incorporate personalized historical dialogues (e.g., UCCR and MemoCRS) achieve notably higher Informativeness than those without historical context. This demonstrates that users' past conversations contain contextual cues that are highly relevant to their current intent, providing specific and semantically rich information that enhances the relevance and interpretability of generated responses. Finally, EyeCRS achieves the best performance on both Fluency and Informativeness, under both human annotation and GPT-4-based automatic evaluation. Unlike models that indiscriminately utilize all historical dialogues, EyeCRS distinguishes and retains personalized information that is strongly aligned with the current context while filtering

Table 2: Subjective evaluation of LLM-based scorer and human annotators for response generation. * denotes statistically significant improvements over best baseline at $p < 0.05$.

	Model	TGRReDial		ReDial	
		Fluency	Informativeness	Fluency	Informativeness
LLM-based scorer	KBRD	1.84	1.80	1.78	1.75
	KGSF	2.05	2.00	1.98	1.95
	UCCR	2.30	2.38	2.30	2.35
	UniCRS	2.28	2.25	2.30	2.19
	ZSCRS	2.60	2.43	2.55	2.40
	MemoCRS	2.70	2.75	2.68	2.72
	EyeCRS	2.80*	2.85*	2.78*	2.82*
Human annotator	KBRD	1.94	1.89	1.87	1.81
	KGSF	2.02	1.98	1.99	1.95
	UCCR	2.35	2.56	2.28	2.55
	UniCRS	2.22	2.18	2.19	2.09
	ZSCRS	2.42	2.23	2.39	2.18
	MemoCRS	2.56	2.62	2.49	2.61
	EyeCRS	2.71*	2.82*	2.67*	2.84*

Table 3: Ablation study on the multi-agent debate module.

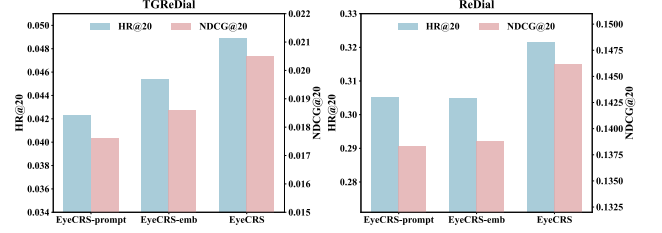
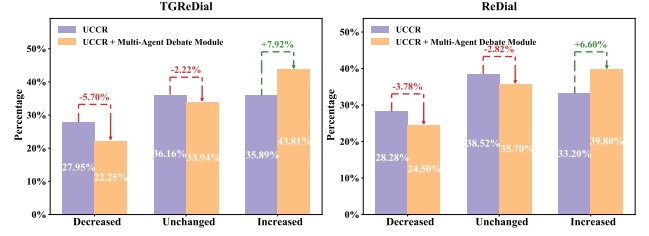
Method	TGRReDial				ReDial			
	HR@10	HR@20	NDCG@10	NDCG@20	HR@10	HR@20	NDCG@10	NDCG@20
w/o debate	0.0253	0.0473	0.0134	0.0189	0.2133	0.3036	0.1158	0.1387
w/o judge	0.0268	0.0471	0.0144	0.0199	0.2167	0.3028	0.1162	0.1381
w/o select	0.0245	0.0456	0.0129	0.0182	0.2023	0.2880	0.1089	0.1356
EyeCRS	0.0282*	0.0489*	0.0151*	0.0205*	0.2273*	0.3215*	0.1225*	0.1461*

out noisy or irrelevant content. This mechanism enables it to focus more precisely on valuable personalized information, resulting in responses with more reasonable reasoning, smoother expression, and richer informational content.

4.3 Ablation Study

Impact of Multi-Agent Debate Module. To evaluate the effectiveness of the multi-agent debate module, we design the following three variants: ‘w/o debate’, which removes the multi-agent debate process and directly relies on the judge role for decision-making. ‘w/o judge’, which replaces the fine-tuned judge role with GPT-4 as the decision maker. ‘w/o select’, which removes both the debate and judge mechanisms, and directly utilizes the user’s entire personalized information from the historical dialogue.

From the results in Table 3, we observe the following: (1) The ‘w/o debate’ variant exhibits a clear performance decline compared with EyeCRS, confirming the effectiveness of the proposed debate mechanism. By engaging supportive and opposing agents in open statements and rebuttals, the model simulates a human-like debate process that facilitates more accurate identification of valuable historical dialogues. (2) The ‘w/o judge’ variant also performs worse than EyeCRS, indicating that our fine-tuned judge, when provided with the same debate reports, exhibits superior decision-making capability compared to a general LLM such as GPT-4. This result suggests that the fine-tuned LLM is better suited to serve as a domain-specific judge, as it more effectively captures task-specific relevance criteria. (3) The ‘w/o select’ variant achieves the lowest performance across both datasets, indicating that indiscriminately utilizing all historical dialogue sessions introduces cognitive negative transfer. This interference disrupts the model’s cognitive learning and decision-making processes, thereby preventing the potential of historical dialogues from being fully leveraged.

**Figure 4: Ablation study on the parametric injection module.****Figure 5: Quantitative analysis of the multi-agent debate module in mitigating cognitive negative transfer.**

Overall, both the debate and judge mechanisms in the multi-agent debate module have been proven effective, facilitating the model in accurately discerning historical dialogue sessions based on the current conversational context.

Impact of Parametric Injection Module. To evaluate the effectiveness of the parametric injection module, we conduct experiments with two variants: (i) ‘EyeCRS-prompt’, which incorporates the useful historical dialogues selected by the multi-agent debate module via prompt-based injection to directly guide the LLM recommender; (ii) ‘EyeCRS-emb’, which encodes these useful historical dialogues into embeddings that are aggregated with the current conversation embeddings before being fed into the recommender.

As shown in Figure 4, both variants underperform compared with EyeCRS. Firstly, ‘EyeCRS-prompt’ suffers from longer input sequences that hinder the model’s ability to process personalized information effectively. More importantly, as previously noted Su et al. [30], Yu and Ananiadou [45], LLMs process external contextual information in a fundamentally different way from their internalized parameter knowledge. This structural difference means that they are less capable of integrating contextual knowledge than parameter information. Consequently, prompt-based injection leads to limited and unstable utilization of historical conversation information. Furthermore, ‘EyeCRS-emb’ also performs worse than our EyeCRS. This variant adopts a static form of information injection, in which the useful historical dialogues are encoded into fixed embeddings and fused with the representations of the current conversation. Such a design limits the model’s capacity for dynamic context reasoning and reduces its flexibility in attending to relevant historical cues across different conversational scenarios.

In contrast, the parametric injection module in EyeCRS internalizes the debate-selected historical information directly into the model’s parameter space, avoiding input expansion and static fusion. This design integrates personalized knowledge as part of the model’s intrinsic decision process, enabling more stable and adaptive use of user-specific information.

Table 4: A sampled case from TGRdial illustrating how historical dialogue impacts recommendation. Green boxes show effective transfer of relevant context, whereas yellow boxes exemplify negative transfer, where misuse of historical dialogue leads to incorrect recommendations.

Current Dialogue	Recommender: Hello there. User: recommend me a <i>comedy</i> .
Historical Dialogues	Session 1: Recommender suggested Joey Wong's Hong Kong films <i>Painted Skin: The Yin Yang Master</i> and <i>A Chinese Ghost Story</i>, also mentioning Tsui Hark and <i>Wicked City</i>. Seeker found these films impressive, especially <i>Painted Skin</i>. Session 2: Recommender mentioned several of Takeshi Kitano's comedies — <i>Boiling Point</i>, <i>Getting Any?</i>, and <i>Violent Cop</i> — along with Bean: <i>The Ultimate Disaster Movie</i>. Seeker said they'd seen them all and found both Kitano's and Bean's blend of humor and warmth impressive. Session 3: Recommender suggested several heartwarming Japanese films — <i>The Recipe for Life After 49 Days</i>, <i>Our Family</i>, and <i>A Farewell to Jinu</i>. Seeker said they'd watch them all, appreciating their warmth, sincerity, and gentle reflections on life and loss.
Response	Standard response: Let me recommend a hilarious comedy — <i>Rat Race</i>. Without historical dialogue sessions: I highly recommend <i>The Wolf of Wall Street</i> — a wild, high-energy black comedy that dives deep into the outrageous excesses of Wall Street. It's both hilarious and scathing, with Leonardo DiCaprio delivering one of his most electrifying performances. The film's sharp dialogue, bold storytelling, and relentless momentum make it an unforgettable and brutally funny look at ambition, greed, and moral chaos. With all historical dialogue sessions: If you enjoy fantasy comedy, I highly recommend watching <i>A Chinese Odyssey Part One: Pandora's Box</i>. It's a hilarious yet touching tale that mixes Stephen Chow's trademark nonsense humor with time travel, romance, and destiny — a perfect pick if you enjoy Hong Kong films. With selected historical dialogue sessions: You've got to check out <i>Rat Race</i> — a wild, fast-paced ensemble comedy where a group of strangers races across the country for a hidden fortune. The cast is stacked, with Rowan Atkinson (<i>Mr. Bean</i> himself!) and other heavy hitters delivering nonstop absurdity and slapstick chaos. What makes it a blast is how it captures that same darkly ironic, offbeat energy you loved in Kitano's <i>Boiling Point</i>. With Atkinson at the center of the madness, it's gleeful mayhem from start to finish — an anarchic ride that perfectly fits your taste.

4.4 Further Analysis

We further validate the effectiveness of EyeCRS in mitigating the negative transfer issue via a quantification analysis and a case study.

Quantification Analysis. To verify the existence of the cognitive negative transfer problem and to evaluate whether our multi-agent debate module can mitigate it, we conduct a quantitative comparative analysis using two model variants: the original UCCR [22] model, which incorporates all historical dialogue sessions, and the extended UCCR with our proposed multi-agent debate module.

As shown in Figure 5,¹ results on the TGRdial dataset reveal that under the original UCCR model, 27.95% of samples are labeled as decreased and 36.16% as unchanged, indicating that incorporating all historical dialogues often introduces irrelevant information and causes cognitive negative transfer. A similar pattern is observed on the ReDial dataset. After integrating the proposed multi-agent debate module into UCCR, the proportion of decreased samples drops by 5.7%, unchanged samples by 2.22%, and increased samples rise by 7.92%. Consistent improvements on both datasets demonstrate that the module effectively filters out distracting dialogue sessions, mitigates negative transfer, and enhances overall accuracy.

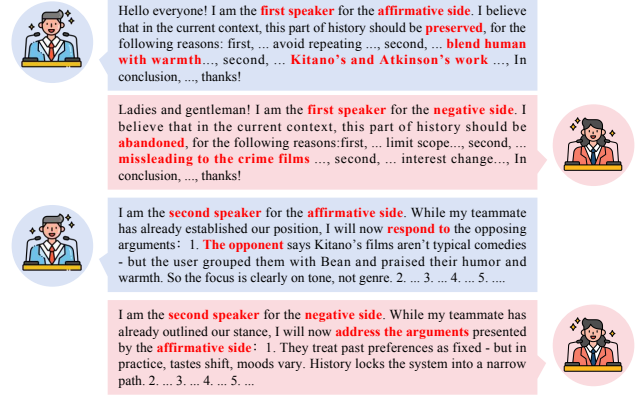


Figure 6: A real debate case study, associated with the current dialogue and the historical dialogue session 2 in Table 4, provides insight into avoiding cognitive negative transfer.

Case Study. To illustrate the impact of historical dialogues on recommendation outcomes, we analyze a representative case from TGRdial (Table 4). The results show that ignoring historical dialogues leads to generic, mismatched recommendations that fail to capture the seeker's humor preference, while indiscriminately using all past sessions introduces irrelevant cues and causes negative transfer, deviating from the seeker's intent. In contrast, selectively retaining only semantically relevant sessions captures fine-grained preferences and yields recommendations better aligned with the seeker's taste. Figure 6 further illustrates how the model distinguishes helpful from harmful historical cues through a debate process, in which explicit reasoning about contextual relevance allows the model to retain the most informative session and avoid negative transfer, leading to more accurate and context-aware recommendations.

5 Conclusion

Inspired by cognitive science, in this work, we identify the issue of cognitive negative transfer in conversational recommendations and propose a novel model, named EyeCRS, to address it. EyeCRS consists of two key modules: a Multi-Agent Debate Module that employs debate among agents to identify and retain beneficial personalized information from historical dialogues and a Parametric Injection Module that parameterizes useful historical dialogue and injects it into an LLM-based recommender, improving its understanding of user interests and recommendation ability. Experiments on two public datasets show that EyeCRS effectively mitigates cognitive negative transfer and achieves superior performance over existing methods. In this paper, we demonstrate the effectiveness of response generation through experiments. However, our work mainly focuses on the recommendation module. Future work will refine the dialogue module to produce more natural, context-aware responses.

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A Prompts in the Main Paper

In this section, we detail the prompts utilized in the multi-agent debate module.

A.1 Statement Stage in the Debate

Task Description: You are to operate as a debate-style intelligent agent who is acting in the specific role of the first speaker. It is your task to present a concise and structured argument regarding the question of whether the given historical dialogue should ultimately be preserved, while explicitly assessing its cognitive impact—namely, by

determining whether it risks negative transfer or whether it provides useful cues for the process of interpreting the user’s current intent.

Evaluation Rules: The evaluation of each historical dialogue considers three key perspectives: semantic continuity, cognitive coherence, and interference controllability. Specifically, we examine whether the historical dialogue remains consistent with the current request in terms of topic, context, or conceptual focus, thereby reducing semantic drift; whether it complements or reinforces the user’s underlying preferences rather than confusing their true interests; and whether it avoids introducing irrelevant or misleading information that could cause cognitive negative transfer.

Response Rules: 1. Follow a debate-style statement consistent with your assigned stance. 2. Reference key phrases from either the historical or current dialogue, enclosed in quotation marks (" "). 3. Keep the response concise (within 100 words), logically coherent, and persuasive.

Input: Stance: {}, Current Conversation: {}, Historical Dialogue: {}.

A.2 Rebuttal Stage in the Debate

Task Description: You are to operate as a debate-style intelligent agent who is acting in the specific capacity of the second speaker. It is your assigned task to deliver a rebuttal that is both concise and structured in response to the prior argument, critically evaluating the claims made by the opponent while simultaneously defending the validity of your own position, with explicit attention devoted to determining whether the historical dialogue causes cognitive negative transfer or whether it offers beneficial cues for the purpose of interpretation.

Evaluation Rules: The evaluation of each rebuttal focuses on the same three key perspectives as in the first stage: semantic continuity, cognitive coherence, and interference controllability. Specifically, we assess whether the rebuttal effectively clarifies semantic alignment with the current request, reinforces consistent user preferences without confusion, and identifies or mitigates possible interference that could cause cognitive negative transfer.

Response Rules: 1. Follow a debate-style rebuttal consistent with your assigned stance. 2. Directly respond to the opponent’s main claims, providing clear counterarguments or refinements. 3. Reference key phrases from either the historical or current dialogue, as well as the previous statements, enclosed in quotation marks (" "). 4. Keep the response concise (within 100 words), logically coherent, and persuasive.

Input: Stance: {}, Current Conversation: {}, Historical Dialogue: {}, Previous Debate Statements: {}.